

7(a)	a (very) small proportion of (alpha) particles are deflected (back) through large angles / angles greater than 90°	B1
	a large proportion of (alpha) particles pass straight through / deflected by small angles	B1
7(b)	electromagnetic wave / electromagnetic radiation	A1
7(c)	neutrino classified as a lepton only <u>and</u> positron classified as a lepton only	B1
	neutron classified as a baryon and a hadron and not as a lepton	B1
7(d)(i)	1.32 u	A1
7(d)(ii)	working states or implies 3 quarks and each quark has a charge of (+) $\frac{2}{3}(e)$	B1
	any combination of 3 quarks comprised of one or more of up / charm / top	B1